Problem 7 – Siege

1 The problem

Let $f\colon S^2 \to S^2$ be a branched cover, $V(f)$ the set of its critical values, and $A, B \subset S^2$ finite sets.

1. If $f(A) \cup V(f) \subset B$, then every complex structure on (S^2, B) can be pulled back to a complex structure on (S^2, A), and this induces the map $f^*\colon \mathrm{Teich}(B) \to \mathrm{Teich}(A)$.
2. If $A \subset B$, then forgetting the points $B \setminus A$ induces the map $i_*\colon \mathrm{Teich}(B) \to \mathrm{Teich}(A)$.

If $f(A) \cup V(f) \cup A \subset B$, then the *deformation space* is defined by

$$\mathcal{D}_f := \{\tau \in \mathrm{Teich}(B) : f^*(\tau) = i_*(\tau)\}.$$

Consider a quadratic rational map f with critical points c_1 and c_2. Assume that c_1 belongs to a cycle of length 4, and the critical value $v = f(c_2)$ does not belong to the cycle. By taking

$$A = \{c_1, f(c_1), f^{\circ 2}(c_1), f^{\circ 3}(c_1)\}, \qquad B = A \cup \{v\},$$

we get a deformation space $\mathcal{D}_f \subset \mathrm{Teich}(B)$. Its image in the moduli space is the set of conjugacy classes of quadratic rational maps with the properties described above.

Explicitly describe a simple closed curve $\gamma \subset (S^2, B)$ such that a power of the Dehn twist around this curve preserves the deformation space, and its action on the set of connected components of $\mathcal{D}_f$ has an infinite orbit.

2 The result

Throughout, f is as in the problem, $a_i = f^i(c_1)$ (indices mod 4), so $A = \{a_0, a_1, a_2, a_3\}$ and $B = A \cup \{v\}$. The critical values of f are $a_1 = f(c_1)$ and v. For an essential simple closed curve $\gamma \subset S^2 \setminus B$ we write T_γ for the Dehn twist about γ, an element of the pure mapping class group $\mathrm{PMod}(S^2, B)$, which acts on $\mathrm{Teich}(B)$ (see §5).

Theorem 1. ***(a) Model case.*** *Let*

$$f = f_{1/4}(z) = 1 - \frac{11}{12z} + \frac{1}{6z^2}.$$

Its 4-cycle is $c_1 = 0 \mapsto \infty \mapsto 1 \mapsto \frac{1}{4} \mapsto 0$, its second critical point is $c_2 = 4/11$, and $v = -25/96$. On $\hat{\mathbb{R}}$ the points of B are ordered as

$$v < c_1 = 0 < f^3(c_1) = \tfrac{1}{4} < f^2(c_1) = 1 < f(c_1) = \infty.$$

Let α be a simple arc from v to $f(c_1) = \infty$ with the following crossing pattern with $\hat{\mathbb{R}}$:

(1) *it leaves v into the lower half-plane and crosses $\mathbb{R}$ upward at a point $x_1 \in (1, \infty)$;*

(2) *it runs in the upper half-plane and crosses $\mathbb{R}$ downward at a point $x_2 \in (0, \frac{1}{4})$;*

(3) *it runs in the lower half-plane and crosses $\mathbb{R}$ upward at a point $x_3 \in (v, 0)$;*

(4) *it stays in the upper half-plane until ∞.*

Thus α meets $\hat{\mathbb{R}}$ only at v, x_1, x_2, x_3, ∞, crossing transversally at x_1, x_2, x_3. A concrete choice is: the lower semicircle from v to 2; then the upper semicircle over $[\frac{1}{8}, 2]$; then the lower semicircle over $[-\frac{1}{8}, \frac{1}{8}]$; then the vertical ray from $-\frac{1}{8}$ up to ∞.

Let $\gamma = \partial N(\alpha)$ be the boundary of a thin closed regular neighbourhood of α. Then γ is a simple closed curve in $S^2 \setminus B$; it bounds a disc containing exactly v and $f(c_1)$, and the complementary disc contains $c_1, f^2(c_1), f^3(c_1)$. The Dehn twist T_γ itself (power 1) satisfies

$$f^* \circ T_\gamma = f^* \qquad \text{and} \qquad i_* \circ T_\gamma = i_*,$$

so $T_\gamma(\mathcal{D}_f) = \mathcal{D}_f$. For every connected component D_0 of $\mathcal{D}_f$ and every $m \neq 0$ we have $T_\gamma^m(D_0) \neq D_0$. Hence every orbit of $\langle T_\gamma \rangle$ on $\pi_0(\mathcal{D}_f)$ is infinite; in particular $\mathcal{D}_f$ has infinitely many components.

(b) Every f. *Let f be arbitrary as in the problem. By Proposition 4.1 there are a unique $t \in W$ and a unique Möbius map M with $M \circ f \circ M^{-1} = f_t$ and $M(c_1) = 0$. Choose a path t_s in W from $\frac{1}{4}$ to t and let h_1 be the endpoint of an isotopy following the marked points (Lemma 11.1). Then the curve*

$$\gamma_f := M^{-1}(h_1(\gamma))$$

has the same property for f: T_{γ_f} preserves $\mathcal{D}_f$, and every orbit of $\langle T_{\gamma_f} \rangle$ on $\pi_0(\mathcal{D}_f)$ is infinite.

When $t \in \left(0, \frac{3-\sqrt{5}}{2}\right)$ this is explicit. The five points of B lie on the circle $C = M^{-1}(\hat{\mathbb{R}})$ in the cyclic order v, c_1, $f^3(c_1)$, $f^2(c_1)$, $f(c_1)$. Call the upper *side of C the image under M^{-1} of the upper half-plane. Then $\gamma_f = \partial N(\alpha_f)$, where α_f is an arc from v to $f(c_1)$ which*

- *leaves v into the lower side and crosses C between $f^2(c_1)$ and $f(c_1)$;*
- *crosses back between c_1 and $f^3(c_1)$;*
- *crosses again between v and c_1;*
- *then runs on the upper side to $f(c_1)$.*

What is known and what is proved here. The setting of the problem, and the disconnectedness of $\mathcal{D}_f$ for quadratic maps with a 4-periodic critical point, go back to E. Hironaka and S. Koch [8]; for background on the pullback map in this generality see [6, 4]. The argument below is self-contained apart from the classical facts (F1)–(F5) listed in §5, which are quoted with references and not reproved. Everything else (the normal form, the fibre formula, the obstruction ($\star$), the linking criterion, the face analysis at $t = \frac{1}{4}$, the isotopy and lifting lemmas) is proved here.

3 Overview of the proof

After a Möbius change of coordinates every f is one of the maps $f_t(z) = 1 + b/z + c/z^2$ with cycle $0 \to \infty \to 1 \to t \to 0$, where t ranges over a connected set W, the complement of nine points in $\mathbb{C}$. The space Teich(A) is the upper half-plane $\mathbb{H}$, with the modular function λ as the universal covering of the moduli space $\mathbb{C} \setminus \{0, 1\}$, so the pair $\Psi = (i_*, f^*)$ maps Teich(B) to $\mathbb{H} \times \mathbb{H}$ and $\mathcal{D}_f = \Psi^{-1}(\text{diagonal})$.

The key algebraic fact is an explicit formula: if a point of Teich(B) has normalised coordinates $(0, \infty, 1, t, v')$, then $w = \lambda(f^*\tau')$ satisfies $V_t(w) = v'$ for an explicit degree-4 rational function V_t.

Since the fifth point v' is never ∞, the value $w = 1-t$ is never attained. Hence $\Psi(\mathrm{Teich}(B))$ avoids the graphs of all elements $g \in \Gamma(2)S$, where $S(\tau) = -1/\tau$; we call this the obstruction $(\star)$. If g has no fixed point in $\mathbb{H}$, the complement X_g of the graph of g in $\mathbb{H} \times \mathbb{H}$ contains the diagonal and has $H_1(X_g) \cong \mathbb{Z}$, measured by winding about the graph. A mapping class T with $\Psi \circ T = \Psi$ preserves $\mathcal{D}_f$. If, moreover, a path from τ to $T^m\tau$ has Ψ-image with nonzero class in $H_1(X_g)$, then τ and $T^m\tau$ cannot lie in the same component of $\mathcal{D}_f$, because a path inside $\mathcal{D}_f$ maps into the contractible diagonal. This is the linking criterion.

For $f_{1/4}$ we take γ around the two critical values v and a_1, along the arc α. Then T_γ is a point push of v, so $i_* \circ T_\gamma = i_*$, and its lift through f is computed from the lift of α through $V_{1/4}$. The map $V_{1/4}$ is Galois with Klein four deck group, and its real locus consists of $\hat{\mathbb{R}}$ and two circles; a face-by-face analysis shows that the lift β of α starting at $w = \frac{1}{4}$ ends at $\frac{3}{4}$ after winding once clockwise around 0. Lifting through λ yields the element $g = T^{-2}S$, which is parabolic and so has no fixed point. From this we get $f^* \circ T_\gamma = f^*$, and the Ψ-image of the fibre path from τ_0 to $T_\gamma^m\tau_0$ winds $\pm m$ times about the graph of g, which proves (a). Part (b) follows by transporting γ along an isotopy that follows the marked points as t moves in W, which conjugates all the structures involved.

4 Normal form

Proposition 4.1. *Put* $f_t(z) = 1 + \dfrac{b}{z} + \dfrac{c}{z^2}$, *with* $b = \dfrac{t^2+t-1}{1-t}$ *and* $c = \dfrac{t(1-2t)}{1-t}$. *Let* E *be the set of* 9 *points*

$$E = \{0, 1, \tfrac{1}{2}\} \cup \{t : t^2 - 3t + 1 = 0\} \cup \{t : t^2 + t - 1 = 0\} \cup \{t : 3t^2 - 3t + 1 = 0\},$$

and put $W = \mathbb{C} \setminus E$. *For* f *as in the problem there is exactly one pair* (t, M) *with* $t \in W$, M *Möbius,* $MfM^{-1} = f_t$ *and* $M(c_1) = 0$. *Conversely, for every* $t \in W$ *the map* f_t *is as in the problem, with cycle* $0 \to \infty \to 1 \to t \to 0$. *Finally,* W *is connected.*

Proof. Existence. Let M be the Möbius map with $M(a_0, a_1, a_2) = (0, \infty, 1)$, put $t = M(a_3)$ and $g = MfM^{-1}$. Then 0 is a critical point of g with $g(0) = \infty$, so 0 is a double pole and the only pole; hence $g = P(z)/z^2$ with $\deg P \le 2$ and $P(0) \neq 0$. Since $g(\infty) = 1$, we get $g = 1 + b/z + c/z^2$ with $c \neq 0$. The conditions $g(1) = t$ and $g(t) = 0$ read $1 + b + c = t$ and $t^2 + bt + c = 0$. Substituting $c = t - 1 - b$ into the second gives $b(t-1) = 1 - t - t^2$. Hence b and c are as stated. (As checks, $b + c = t - 1$ and $(1-t)(t^2 + bt + c) = 0$ identically.)

Uniqueness. Any other such M' maps $c_1 = a_0$ to 0, and then a_1, a_2 to $f_{t'}(0) = \infty$ and $f_{t'}(\infty) = 1$; so $M'(a_0, a_1, a_2) = (0, \infty, 1)$ and $M' = M$.

Which t *occur.* The value $t = \frac{1}{2}$ gives $c = 0$, and the degree drops. For $t \notin \{0, 1, \frac{1}{2}\}$ the map f_t has degree 2 and 0 is a critical point of exact period 4, since $0, \infty, 1, t$ are distinct. Since $f_t'(z) = -(bz + 2c)/z^3$, we have $c_2 = -2c/b$, or $c_2 = \infty = a_1$ if $b = 0$, i.e. if $t^2 + t - 1 = 0$. Otherwise

$$v(t) = 1 - \frac{b^2}{4c} = -\frac{(t^2 - 3t + 1)^2}{4t(1-t)(1-2t)},$$

using $(t^2 + t - 1)^2 - (t^2 - 3t + 1)^2 = 4t(1-t)(1-2t)$. Similarly, with $D = 4t(1-t)(1-2t)$, one has $v - 1 = -(t^2 + t - 1)^2/D$ and $v - t = -(3t^2 - 3t + 1)^2/D$; the latter because

$$(t^2 - 3t + 1)^2 + t\,D = 9t^4 - 18t^3 + 15t^2 - 6t + 1 = (3t^2 - 3t + 1)^2.$$

Hence $v = 0 \iff t^2 - 3t + 1 = 0$, $v = 1 \iff b = 0$, and $v = t \iff 3t^2 - 3t + 1 = 0$. So W is exactly the set of admissible parameters. It is $\mathbb{C}$ minus finitely many points, hence connected. $\square$

At $t = \frac{1}{4}$ one gets $b = -\frac{11}{12}$, $c = \frac{1}{6}$, $c_2 = \frac{4}{11}$ and $v = -\frac{25}{96}$, which gives the map $f_{1/4}$ of Theorem 1(a).

Factorisation. Put $N_t(z) = z/(z - c_2(t))$ and $M_t(y) = v(t) + (1 - v(t))/y$. Then

$$f_t(z) = v + (1-v)\Big(1 - \frac{c_2}{z}\Big)^2 = M_t\big(N_t(z)^2\big). \tag{1}$$

To check it, compare coefficients: $-2(1-v)c_2 = b$ and $(1-v)c_2^2 = c$, which follow from $1-v = b^2/(4c)$ and $c_2 = -2c/b$. Both M_t and N_t depend continuously on $t \in W$. Also $N_t(0) = 0$, $N_t(c_2) = \infty$, $M_t(0) = \infty$ and $M_t(\infty) = v$.

5 Teichmüller preliminaries

Conventions. A marking of (S^2, X), with $|X| \geq 3$, is an orientation-preserving homeomorphism $\varphi\colon S^2 \to \hat{\mathbb{C}}$. $\mathrm{Teich}(X)$ is the set of markings modulo the relation $\varphi_1 \sim \varphi_2$, which holds iff there is a Möbius map M with $M\varphi_1 = \varphi_2$ on X and $M\varphi_1 \simeq \varphi_2$ rel X. The pure mapping class group $\mathrm{PMod}(S^2, X)$ acts by $g\cdot[\varphi] = [\varphi g^{-1}]$; for a mapping class T we write $F \circ T$ for the map $\tau \mapsto F(T\cdot\tau)$. The pullback $f^*[\varphi] = [\psi]$ is defined by requiring $\varphi \circ f \circ \psi^{-1}$ to be rational. In our situation $S^2 = \hat{\mathbb{C}}$, and $\tau_0 = [\mathrm{id}]$ denotes the base point.

Standard facts quoted, not reproved.

(F1) (Modular function; classical, see e.g. [1, 5].) Let $\lambda = (\theta_2/\theta_3)^4\colon \mathbb{H} \to \mathbb{C} \setminus \{0, 1\}$. It is a holomorphic universal covering whose fibres are the orbits of $\Gamma(2)$. Moreover $\lambda(\tau + 2) = \lambda(\tau)$ and $\lambda(-1/\tau) = 1 - \lambda(\tau)$; $\lambda(\tau) = 16e^{i\pi\tau} + O(e^{2\pi i\tau})$ as $\operatorname{Im}\tau \to \infty$; and $y \mapsto \lambda(iy)$ is a decreasing bijection $(0, \infty) \to (0, 1)$.

(F2) (Teichmüller; see e.g. [9].) $\mathrm{Teich}(X)$ is a complex manifold homeomorphic to $\mathbb{R}^{2(|X|-3)}$; in particular it is contractible. $\mathrm{PMod}(S^2, X)$ acts properly discontinuously by biholomorphisms, and the quotient is the moduli space $\mathcal{M}_{0,X}$.

(F3) (Bers fibration and Birman exact sequence [2, 3, 7].) The forgetful map $i_*\colon \mathrm{Teich}(B) \to \mathrm{Teich}(A)$ is surjective. Fix $p \in \mathrm{Teich}(A)$ and normalise a_0, a_1, a_2, a_3 to $0, \infty, 1, t$; write $A_p = \{0, \infty, 1, t\}$. Recording the position v' of the fifth point identifies the fibre $i_*^{-1}(p)$ with the universal cover of $\hat{\mathbb{C}} \setminus A_p$. The point-push subgroup $\ker\big(\mathrm{PMod}(S^2, B) \to \mathrm{PMod}(S^2, A)\big) \cong \pi_1(S^2 \setminus A, v)$, $[\ell] \mapsto \mathrm{Push}(\ell)$, acts on each fibre as the deck group: for the fibre through τ_0, $\mathrm{Push}(\ell)^{\pm 1} \cdot \tau_0$ is the endpoint of the lift through v' of ℓ starting at τ_0, the sign depending only on conventions.

(F4) (Push formula, see [7].) The push of v along a simple loop δ is $T_{\delta_L} T_{\delta_R}^{-1}$, where δ_L, δ_R are the two boundary curves of a regular neighbourhood of δ.

(F5) (Douady–Hubbard [6].) f^* is well defined and continuous; indeed it is holomorphic. (In [6] this is proved for Thurston's pullback map; the proof applies verbatim to $f^*\colon \mathrm{Teich}(B) \to \mathrm{Teich}(A)$.)

The push formula is used only up to sign, and every argument below is symmetric under $m \mapsto -m$.

Lemma 5.1 (free action)**.** $\mathrm{PMod}(S^2, X)$ *acts freely on* $\mathrm{Teich}(X)$.

Proof. Suppose $[\varphi g^{-1}] = [\varphi]$. Then $M\varphi = \varphi g^{-1}$ on X for some Möbius M. Since g fixes X pointwise, M fixes the at least three points of $\varphi(X)$, so $M = \mathrm{id}$. Then $\varphi g^{-1} \simeq \varphi$ rel X, so $g = 1$ in $\mathrm{PMod}(S^2, X)$. □

By **(F2)** and Lemma 5.1, the modulus map $\mathrm{mod}\colon \mathrm{Teich}(A) \to \mathbb{C} \setminus \{0,1\}$ is a covering, and since $\mathrm{Teich}(A)$ is simply connected it is a universal covering. It sends $[\varphi]$ to the t for which some Möbius map takes $\varphi(a_0, a_1, a_2, a_3)$ to $(0, \infty, 1, t)$. By **(F1)** there is a biholomorphism $\Theta\colon \mathrm{Teich}(A) \to \mathbb{H}$ with $\lambda \circ \Theta = \mathrm{mod}$, and we use it to identify $\mathrm{Teich}(A)$ with $\mathbb{H}$. For $f = f_{1/4}$ the base point $\tau_0 = [\mathrm{id}] \in \mathrm{Teich}(B)$ lies in $\mathcal{D}_f$, because $f^*[\mathrm{id}] = [\mathrm{id}]$, f being rational. Composing Θ with an element of $\Gamma(2)$ we normalise it so that $i_*\tau_0 = p_0 := iy_0$, where $\lambda(iy_0) = \frac{1}{4}$ (numerically $y_0 \approx 1.2793$).

Put $\Psi = (i_*, f^*)\colon \mathrm{Teich}(B) \to \mathbb{H} \times \mathbb{H}$. Then $\mathcal{D}_f = \Psi^{-1}(\Delta)$, where Δ is the diagonal.

Lemma 5.2 (equivariance for a lift fixing A)**.** *Let φ be a homeomorphism fixing B pointwise, and $\tilde\varphi$ a homeomorphism fixing A pointwise with $f\tilde\varphi = \varphi f$. Then $f^*(\varphi \cdot \tau) = \tilde\varphi \cdot f^*(\tau)$.*

Proof. Let $\tau = [\psi_0]$ and suppose $\psi_0 f = F\psi$ with F rational. Then $\psi_0\varphi^{-1} f = \psi_0 f \tilde\varphi^{-1} = F(\psi\tilde\varphi^{-1})$, so $f^*[\psi_0\varphi^{-1}] = [\psi\tilde\varphi^{-1}]$. □

A lift of an element of $\mathrm{PMod}(S^2, B)$ need not fix A pointwise: it may exchange a_i with the other preimage of a_{i+1}. So Lemma 5.2 has to be applied to an explicitly constructed lift, as in the next lemma.

Lemma 5.3 (lift of T_γ)**.** *Let α be a simple arc from v to a_1 with interior in $S^2 \setminus B$. Let $\gamma = \partial N(\alpha)$, and realise T_γ in a collar $R \subset S^2 \setminus N(\alpha)$ disjoint from B. Then:*

(i) *$f^{-1}(R) = R_1 \sqcup R_2$, with each R_j mapped homeomorphically onto R;*

(ii) *the map $\tilde T$ defined as $(f|_{R_j})^{-1} T_\gamma f$ on R_j and the identity elsewhere is a homeomorphism with $f\tilde T = T_\gamma f$;*

(iii) *$\tilde T$ fixes $f^{-1}(B) \supset A$ pointwise.*

Hence $f^* \circ T_\gamma = [\tilde T] \cdot f^*$ with $[\tilde T] \in \mathrm{PMod}(S^2, A)$.

Proof. R contains no critical value, so $f^{-1}(R) \to R$ is a 2-sheeted covering. Its monodromy around the core of R is the product of the monodromies around the two critical values v, a_1 inside γ. Each of these is the transposition of the two sheets, so the product is trivial. Hence the covering splits into two annuli, each mapped homeomorphically. Since $T_\gamma = \mathrm{id}$ on ∂R, $\tilde T$ is continuous and bijective, hence a homeomorphism of S^2, and $f\tilde T = T_\gamma f$ holds on each R_j and trivially elsewhere. The set $f^{-1}(B)$ is disjoint from $f^{-1}(R)$, so $\tilde T$ fixes it; it contains A because $f(A) = A \subset B$. Lemma 5.2 gives the last statement. □

6 The fibre formula and the obstruction $(\star)$

Lemma 6.1. *Let $\tau' \in \mathrm{Teich}(B)$ be represented, after normalising, by*

$$(\hat{\mathbb{C}}; a_0, a_1, a_2, a_3, v) = (0, \infty, 1, t, v').$$

Then $w := \lambda(f^*\tau')$ satisfies

$$V_t(w) = v', \qquad \text{where} \quad V_t(w) = -\frac{\big((w-1)^2 - t\big)^2}{4w(w-1)(w+t-1)}.$$

Proof. The pulled-back structure carries a degree-2 rational map F whose critical values are the images ∞ and v' of the critical values a_1 and v. Up to a Möbius change in the source, $F(z) = v' + z^2$. Since $a_0 = c_1$ is the critical point over a_1, its marked image is $a'_0 = \infty$. The other points satisfy $F(a'_i) =$ (image of a_{i+1}), so $a'_1 = s_2$, $a'_2 = s_3$, $a'_3 = s_0$ with

$$s_2^2 = 1 - v', \qquad s_3^2 = t - v', \qquad s_0^2 = -v';$$

the marking determines the signs. The map $z \mapsto (s_3 - s_2)/(z - s_2)$ sends (∞, s_2, s_3, s_0) to $(0, \infty, 1, w)$, so $w = (s_3 - s_2)/(s_0 - s_2)$.

Put $u = s_2 - s_0$ and $U = u^2$. From $s_2^2 - s_0^2 = 1$ we get $s_2 + s_0 = u^{-1}$, so $s_2 = (u + u^{-1})/2$ and $s_0 = (u^{-1} - u)/2$. Also $s_3 = s_2 - wu$. The relation $s_3^2 - s_0^2 = t$ becomes $1 - w(U+1) + w^2 U = t$, so

$$U = \frac{w + t - 1}{w(w-1)}, \qquad 1 - U = \frac{(w-1)^2 - t}{w(w-1)}.$$

Then $v' = -s_0^2 = -(1-U)^2/(4U)$, which is the stated formula. □

Note that $V_t(t) = -\dfrac{(t^2 - 3t + 1)^2}{4t(t-1)(2t-1)} = v(t)$.

Corollary 6.2 (the obstruction $(\star)$)**.** *$V_t^{-1}(\infty) = \{0, 1, \infty, 1-t\}$, and at $w = 1 - t$ the numerator of V_t equals $(t^2 - t)^2 \neq 0$. Since $v' \neq a_1 = \infty$, we get $w \neq 1 - t$. Thus*

$$\lambda(f^*\tau') \neq 1 - \lambda(i_*\tau') \qquad \text{for all } \tau' \in \mathrm{Teich}(B). \tag{$\star$}$$

*By **(F1)**, $\lambda(q) = 1 - \lambda(p) = \lambda(-1/p)$ holds iff $q = g(p)$ for some $g \in \Gamma(2)S$, where $S(\tau) = -1/\tau$. So $\Psi(\mathrm{Teich}(B))$ avoids every graph $\Gamma_g = \{(p, g(p)) : p \in \mathbb{H}\}$ with $g \in \Gamma(2)S$.*

Proof. The denominator vanishes at $0, 1, 1-t$, which are distinct since $t \neq 0, 1$, and the numerator is $(1-t)^2$, t^2, $(t^2 - t)^2$ there, all nonzero. Since the numerator has degree 4 and the denominator degree 3, also $V_t(\infty) = \infty$. The rest was explained in the statement, using $\lambda(i_*\tau') = t$. □

Lemma 6.3 (branching of V_t)**.** *For $t \in W$, V_t has degree 4 and is unbranched over $\hat{\mathbb{C}} \setminus \{0, 1, t\}$.*

Proof. By the proof of Corollary 6.2, numerator and denominator are coprime, so $\deg V_t = 4$. Fix $v' \notin \{0, 1, t, \infty\}$. If $V_t(w) = v'$, then $w \notin \{0, 1, 1-t, \infty\}$; reversing the computation in the proof of Lemma 6.1 (choose u with $u^2 = U$ and define s_2, s_0, s_3 by the formulas there) shows that $w = (s_3 - s_2)/(s_0 - s_2)$ for some choice of square roots s_2, s_3, s_0 of $1 - v', t - v', -v'$. So the solutions w come from the sign choices of (s_2, s_3, s_0) modulo a global sign, and there are at most 4 of them.

They are distinct. If two sign classes gave the same w, the Möbius map carrying one quadruple (∞, s_2, s_3, s_0) to the other would be an affine map L with $L(s_j) = \pm s_j$, with a sign pattern that is not constant. Then $L(z)^2 - z^2$, of degree at most 2, would vanish at the three distinct points s_2, s_3, s_0, forcing $L = \pm\mathrm{id}$, a contradiction. Over ∞ there are 4 simple preimages $0, 1, 1-t, \infty$. Hence every point outside $\{0, 1, t\}$ has 4 distinct preimages. □

Lemma 6.4 (fibre lemma)**.** *Fix $p \in \mathrm{Teich}(A)$ with $\lambda(p) = t$, and let $\tilde{U} = i_*^{-1}(p)$. By **(F3)**, $\tilde{U}$ is the universal cover of $\hat{\mathbb{C}} \setminus A_p$ via $\tau' \mapsto v'(\tau')$. Then $w = \lambda \circ f^*$ is the lift of v' through the covering $V_t \colon \hat{\mathbb{C}} \setminus V_t^{-1}(A_p) \to \hat{\mathbb{C}} \setminus A_p$, and $f^*|_{\tilde{U}}$ is the lift of w through λ. Both lifts are determined by their value at one point.*

Proof. By Lemma 6.3, V_t restricted as stated is a covering, since all its critical values lie in A_p. By **(F5)** and Lemma 6.1, $w = \lambda \circ f^*$ is continuous on $\tilde{U}$ and satisfies $V_t \circ w = v'$; as $v' \notin A_p$, w takes values in $\hat{\mathbb{C}} \setminus V_t^{-1}(A_p)$. So w is a lift of v', and $f^*|_{\tilde{U}}$ is a lift of w through λ by definition. Since $\tilde{U}$ is connected, lifts are determined by their value at one point. □

7 $\mathcal{D}_f$ is a smooth curve

Lemma 7.1. *$\mathcal{D}_f$ is a one-dimensional complex submanifold of* $\mathrm{Teich}(B)$. *Hence its connected components are path components.*

Proof. The map $\mathrm{Teich}(B) \to \mathcal{M}_{0,5}$, $\tau' \mapsto (t', v')$, is a covering by **(F2)** and Lemma 5.1, so (t', v') are local coordinates. Let $\tau_1 \in \mathcal{D}_f$ have coordinates (t_1, v_1). Then $w(\tau_1) = \lambda(f^*\tau_1) = \lambda(i_*\tau_1) = t_1$, so $v_1 = V_{t_1}(t_1) = v(t_1)$. Since $v_1 \notin \{0, 1, t_1, \infty\}$, Lemma 6.3 makes V_{t_1} a local biholomorphism at $w = t_1$. By the implicit function theorem and continuity of w, near τ_1 the function $w(t', v')$ is the holomorphic branch of $V_{t'}^{-1}(v')$ through t_1. Moreover λ is injective near $i_*\tau_1 = f^*\tau_1$. So near τ_1 the condition $f^*\tau' = i_*\tau'$ is equivalent to $w(t', v') = t'$, i.e. to $v' = V_{t'}(t') = v(t')$, which defines a holomorphic graph. □

8 The linking criterion

Proposition 8.1. *Let $T \in \mathrm{PMod}(S^2, B)$ satisfy $\Psi \circ T = \Psi$, so that $T^{\pm 1}(\mathcal{D}_f) = \mathcal{D}_f$. Let $g \in \Gamma(2)S$ have no fixed point in $\mathbb{H}$. Let $\tau \in \mathcal{D}_f$ with $p_0 = i_*\tau$, and let D_0 be its component. Suppose that for each $m \neq 0$ there is a path σ'_m in the fibre $i_*^{-1}(p_0)$ from τ to $T^m\tau$ such that the loop $f^* \circ \sigma'_m$ has nonzero winding number about $g(p_0)$. Then $T^m D_0 \neq D_0$ for every $m \neq 0$.*

Proof. Let $X_g = \mathbb{H} \times \mathbb{H} \setminus \Gamma_g$. The map $(p, q) \mapsto \big(p, (q - \operatorname{Re} g(p))/\operatorname{Im} g(p)\big)$ identifies X_g with $\mathbb{H} \times (\mathbb{H} \setminus \{i\})$, and is an orientation-preserving affine map on each fibre $\{p\} \times \mathbb{H}$. So $H_1(X_g) \cong \mathbb{Z}$, and the class of a loop in $\{p\} \times \mathbb{H}$ is its winding number about $g(p)$. By $(\star)$, $\Psi(\mathrm{Teich}(B)) \subset X_g$, and since g has no fixed point, $\Delta \subset X_g$.

Suppose $T^m D_0 = D_0$. By Lemma 7.1 there is a path $\sigma \subset D_0$ from τ to $T^m\tau$. Since $\mathrm{Teich}(B)$ is contractible **(F2)**, $\sigma \simeq \sigma'_m$ rel endpoints. Applying Ψ, the loops $\Psi\sigma$ and $\Psi\sigma'_m$ (they are loops because $\Psi(T^m\tau) = \Psi(\tau)$) are homotopic in X_g. But $\Psi\sigma \subset \Delta$, which is contractible, so $\Psi\sigma$ is null-homologous in X_g, whereas $\Psi\sigma'_m = (p_0, f^*\sigma'_m)$ has nonzero class. This is a contradiction. □

9 The computation at $t_0 = \frac{1}{4}$

From now on $f = f_{1/4}$ and $V = V_{1/4}$. Since $(w-1)^2 - \frac{1}{4} = (w - \frac{1}{2})(w - \frac{3}{2})$,

$$V(w) = -\frac{(w - \frac{1}{2})^2 (w - \frac{3}{2})^2}{4w(w-1)(w - \frac{3}{4})}.$$

Deck group. Put $g_0(w) = \dfrac{w - 3/4}{w - 1}$, $g_2(w) = \dfrac{3}{4w}$ and $g_3 = g_0 g_2 = g_2 g_0 = \dfrac{3(w-1)}{4(w - 3/4)}$; g_0 and g_2 are commuting involutions. Direct substitution gives $V \circ g_0 = V \circ g_2 = V$:

- under g_2: $w - \frac{1}{2} \mapsto -(w - \frac{3}{2})/(2w)$, $w - \frac{3}{2} \mapsto -3(w - \frac{1}{2})/(2w)$, and $4w(w-1)(w - \frac{3}{4}) \mapsto 9(w - \frac{3}{4})(w-1)/(4w^3)$;
- under g_0: $w - \frac{1}{2} \mapsto (w - \frac{1}{2})/(2(w-1))$, $w - \frac{3}{2} \mapsto -(w - \frac{3}{2})/(2(w-1))$, and the denominator becomes $w(w - \frac{3}{4})/(4(w-1)^3)$.

So V is Galois with Klein four deck group $\{\mathrm{id}, g_0, g_2, g_3\}$. Its 6 critical points are the fixed points of g_0, g_2, g_3: $\frac{1}{2}, \frac{3}{2}$, which lie over 0; $\pm\frac{\sqrt{3}}{2}$, which lie over 1; and $P, \bar{P} = \frac{3}{4} \pm \frac{\sqrt{3}}{4} i$, which lie over $\frac{1}{4}$. The critical values are therefore $0, 1, \frac{1}{4} \in A$.

Real locus. V has real coefficients, so $V(w) \in \hat{\mathbb{R}}$ iff $V(\bar{w}) = V(w)$ iff $\bar{w} = h(w)$ for some deck map h: $h = \mathrm{id}$ gives $\hat{\mathbb{R}}$; $h = g_0$ gives the circle $C_1 \colon |w-1| = \frac{1}{2}$; $h = g_2$ gives $C_2 \colon |w| = \frac{\sqrt{3}}{2}$; and $h = g_3$ gives $|w - \frac{3}{4}|^2 = -\frac{3}{16}$, which is empty. Moreover $C_1 \cap C_2 = \{P, \bar{P}\}$.

Faces. All critical values of V are real. So V restricted to $\hat{\mathbb{C}} \setminus V^{-1}(\hat{\mathbb{R}})$ is a covering of the two open half-planes, and each complementary component (face) maps homeomorphically onto one half-plane. Adjacent faces map to opposite half-planes, because V is a local homeomorphism along each edge. We need four faces, all inside C_2 and hence avoiding $w = 1$:

$$L_2 = \{\operatorname{Im} w < 0,\ |w| < \tfrac{\sqrt{3}}{2},\ |w-1| > \tfrac{1}{2}\}, \qquad U_2 = \overline{L_2} \text{ (complex conjugate)},$$
$$U_{12} = \{\operatorname{Im} w > 0,\ |w| < \tfrac{\sqrt{3}}{2},\ |w-1| < \tfrac{1}{2}\}, \qquad L_{12} = \overline{U_{12}}.$$

Boundary values. The vertices of the graph $V^{-1}(\hat{\mathbb{R}})$ are the points of $V^{-1}(\{0, 1, \frac{1}{4}, \infty\}) = \{\frac{1}{2}, \frac{3}{2}, \pm\frac{\sqrt{3}}{2}, P, \bar{P}, 0, 1, \frac{3}{4}, \infty\}$. Each edge contains no vertex except at its ends, so it maps homeomorphically onto the arc of $\hat{\mathbb{R}}$ between the images of its ends that avoids the other marked values.

- On $(-\frac{\sqrt{3}}{2}, 0)$, V increases from 1 to $+\infty$: here $V \approx -3/(16w)$ near 0, and $V(-\frac{\sqrt{3}}{2}) = 1$.
- On $(0, \frac{1}{2})$, V increases from $-\infty$ to 0, with $V(\frac{1}{4}) = v$.
- On $(\frac{1}{2}, \frac{3}{4})$, V decreases from 0 to $-\infty$, with $V(\frac{2}{3}) = v$, since $\frac{2}{3} = g_0(\frac{1}{4})$.
- The C_1-arcs from $\frac{1}{2}$ to P and from $\frac{1}{2}$ to $\bar{P}$ map onto $(0, \frac{1}{4})$: the alternative arc of $\hat{\mathbb{R}}$ from 0 to $\frac{1}{4}$ would pass through ∞ and 1, whose preimages $0, 1, \frac{3}{4}, \pm\frac{\sqrt{3}}{2}$ are not on these arcs.
- The C_2-arcs from P to $-\frac{\sqrt{3}}{2}$ (outside C_1) and from P to $\frac{\sqrt{3}}{2}$ (inside C_1), and their conjugates, map onto $(\frac{1}{4}, 1)$; the segment $(\frac{3}{4}, \frac{\sqrt{3}}{2})$ maps onto $(1, \infty)$.
- $V'(\frac{1}{4}) > 0$ and V is real on $\mathbb{R}$, so U_2, which lies just above $\frac{1}{4}$, maps to the upper half-plane.

Since U_2 is adjacent to L_2 across $(-\frac{\sqrt{3}}{2}, 0)$ and to U_{12} across the C_1-arc from $\frac{1}{2}$ to P, and U_{12} is adjacent to L_{12} across $(\frac{1}{2}, \frac{3}{4})$, it follows that U_2 and L_{12} map to the upper half-plane, and L_2 and U_{12} to the lower.

Lift of α. Let β be the lift of α through V starting at $w = \frac{1}{4}$, which lies over v. The four sub-arcs of α between consecutive points of $\alpha \cap \hat{\mathbb{R}}$ each lie in one open half-plane, and on each face the inverse of V is a homeomorphism onto the closure of that face; at the crossing points V is a local biholomorphism, real on $\mathbb{R}$.

1. α leaves v downward. Since $V'(\frac{1}{4}) > 0$, β enters L_2. On ∂L_2 the preimage of $(1, \infty)$ is $(-\frac{\sqrt{3}}{2}, 0)$, where V is increasing, so β crosses $\mathbb{R}$ there, upward, into U_2.
2. On ∂U_2 the preimage of $(0, \frac{1}{4})$ is the C_1-arc from $\frac{1}{2}$ to P. β crosses it into U_{12}.
3. On ∂U_{12} the preimage of $(v, 0)$ is $(\frac{1}{2}, \frac{2}{3})$, where V is decreasing. β crosses it downward into L_{12}.
4. The only preimage of ∞ in $\overline{L_{12}}$ is $\frac{3}{4}$ (the poles 0 and 1 lie outside C_1 and C_2 respectively). So β ends at $w = \frac{3}{4} = 1 - t_0$.

In particular only the crossing pattern of α was used, not the concrete realisation. The arc β lies in the disc $|w| < \frac{\sqrt{3}}{2}$, which avoids 1, and avoids 0. It meets $\mathbb{R}$ only at $\frac{1}{4}$, at one point of the negative real axis (crossed upward), at one point of $(\frac{1}{2}, \frac{2}{3})$ (crossed downward), and at $\frac{3}{4}$, which it reaches from below. Following the argument of w along β followed by the segment $[\frac{3}{4}, \frac{1}{4}]$, it decreases by 2π. So this closed loop has winding number -1 about 0, and since the homotopy class of a loop in a punctured disc is determined by its winding number, in $\mathbb{C} \setminus \{0, 1\}$

$$\beta \simeq (\text{clockwise loop about 0 based at } \tfrac{1}{4}) \cdot [\tfrac{1}{4}, \tfrac{3}{4}] \qquad \text{rel endpoints.} \tag{2}$$

Lift through λ. The clockwise loop about 0 is homotopic to: go from $\frac{1}{4}$ to ε along $\mathbb{R}$, around the small clockwise circle $|w| = \varepsilon$, and back. By **(F1)** its λ-lift from $p_0 = iy_0$ goes up the imaginary axis, then decreases $\operatorname{Re}\tau$ by 2 (as $\lambda \approx 16e^{i\pi\tau}$ for large $\operatorname{Im}\tau$), and comes back down; it ends at $p_0 - 2$. By $\lambda(-1/\tau) = 1 - \lambda(\tau)$ and $\lambda(\tau - 2) = \lambda(\tau)$, the segment $[\frac{1}{4}, \frac{3}{4}]$ then lifts to the vertical segment from $p_0 - 2$ to $-2 + i/y_0$. So the endpoint of the λ-lift of β from p_0 is

$$q^* = g(p_0), \qquad g = T^{-2}S = \begin{pmatrix} -2 & -1 \\ 1 & 0 \end{pmatrix} \in \Gamma(2)S, \qquad \operatorname{tr} g = -2,$$

where T here denotes $\tau \mapsto \tau + 1$. So g is parabolic and has no fixed point in $\mathbb{H}$.

Remark 9.1 (orientation conventions)**.** Had the winding of β about 0 been $+1$ instead of -1 (e.g. under the opposite orientation convention), the same computation would give $g = T^2S = \begin{pmatrix} 2 & -1 \\ 1 & 0 \end{pmatrix}$, of trace 2, again parabolic. More generally $T^{2k}S$ has trace $2k$ and is fixed-point-free on $\mathbb{H}$ for every $k \neq 0$; only the fact that the winding is nonzero matters.

10 Proof of Theorem 1(a)

α is simple and γ is as claimed. In the concrete realisation, the lower semicircle from v to 2 has centre ≈ 0.870 and radius ≈ 1.130; the small lower semicircle over $[-\frac{1}{8}, \frac{1}{8}]$ lies within distance 0.995 of that centre, so strictly inside it. The upper semicircle over $[\frac{1}{8}, 2]$ has real parts in $[\frac{1}{8}, 2]$, so it misses the vertical ray at $\operatorname{Re} z = -\frac{1}{8}$. The arc meets $\hat{\mathbb{R}}$ only at $v, 2, \frac{1}{8}, -\frac{1}{8}, \infty$, with $2 \in (1, \infty)$, $\frac{1}{8} \in (0, \frac{1}{4})$ and $-\frac{1}{8} \in (v, 0)$, crossing transversally in the required directions. So α is simple, has the crossing pattern, and meets B only at its ends. For any simple arc α with the crossing pattern, $\gamma = \partial N(\alpha)$ bounds a disc containing exactly v, a_1, because α is simple with interior in $S^2 \setminus B$; and the lifting argument of §9 used only the crossing pattern. Everything below therefore applies to any such arc.

T_γ is a point push. Let ℓ be the lasso: follow α to near a_1, go once around a_1, and return. A regular neighbourhood of ℓ is an annulus. One of its boundary curves bounds a disc containing only a_1 (so the twist about it is trivial), and the other is γ. So by **(F4)**, $\operatorname{Push}(\ell) = T_\gamma^{\pm 1}$. In particular T_γ maps to 1 in $\operatorname{PMod}(S^2, A)$, so $i_* \circ T_\gamma = i_*$.

T_γ preserves $\mathcal{D}_f$. Work in the fibre $i_*^{-1}(p_0)$ through τ_0; here $A_{p_0} = \{0, \infty, 1, \frac{1}{4}\}$, $v'(\tau_0) = v$, $w(\tau_0) = \lambda(f^*\tau_0) = \frac{1}{4}$ and $f^*\tau_0 = p_0$, because $\tau_0 \in \mathcal{D}_f$.

- By **(F3)**, there is a path σ' from τ_0 to $T_\gamma^{\pm 1}\tau_0$ which projects to ℓ under v'.
- Truncate α to α', stopping just before a_1. By the fibre lemma 6.4, $w \circ \sigma'$ is the V-lift of ℓ starting at $\frac{1}{4}$, which is $\beta' \cdot c \cdot \beta'^{-1}$. Here β' is the corresponding truncation of β, and c is a small loop about $\frac{3}{4}$; this uses that $\frac{3}{4}$ is a simple preimage of ∞.
- Since $\frac{3}{4} \notin \{0, 1\}$, c is null-homotopic in $\mathbb{C} \setminus \{0, 1\}$. So its λ-lift is a small closed loop $\tilde{c}$ about q^* (recall $\lambda(q^*) = \frac{3}{4}$ and λ is a local biholomorphism), and $f^*\sigma' = \tilde{\beta}' \cdot \tilde{c} \cdot \tilde{\beta}'^{-1}$ is closed, where $\tilde{\beta}'$ is the λ-lift of β' from p_0.
- Hence $f^*(T_\gamma^{\pm 1}\tau_0) = f^*(\tau_0)$. By Lemma 5.3, $[\tilde{T}]^{\pm 1}$ fixes the point $f^*\tau_0$, so $[\tilde{T}] = 1$ by Lemma 5.1.
- Therefore $f^* \circ T_\gamma = f^*$ on all of $\operatorname{Teich}(B)$. Together with $i_* \circ T_\gamma = i_*$ this gives $\Psi \circ T_\gamma = \Psi$, and so $T_\gamma^{\pm 1}(\mathcal{D}_f) = \mathcal{D}_f$.

Infinite orbits. Fix $m \neq 0$, and let σ'_m be the path in the fibre from τ_0 to $T_\gamma^m \tau_0$ lying over the loop $\ell^{\pm m}$ (**(F3)**). As above, $f^*\sigma'_m = \tilde{\beta}'\,\tilde{c}^{\pm m}\,\tilde{\beta}'^{-1}$. This loop lies in $\mathbb{H} \setminus \{g(p_0)\}$ by $(\star)$, and its winding number about $q^* = g(p_0)$ is $\pm m \neq 0$, because $\tilde{c}$ winds once about q^* (it is the image of a small circle about $\frac{3}{4}$ under a local inverse of λ). So the class of $\Psi \circ \sigma'_m$ in $H_1(X_g) \cong \mathbb{Z}$ is $\pm m$, and Proposition 8.1 (with $T = T_\gamma$, $\tau = \tau_0$) gives $T_\gamma^m D_0 \neq D_0$ for the component D_0 of τ_0.

Now let $\tau_1 \in \mathcal{D}_f$ be arbitrary and D_1 its component. Choose a path ρ from τ_0 to τ_1. Then $\rho^{-1} \cdot \sigma'_m \cdot T_\gamma^m \rho$ is a path from τ_1 to $T_\gamma^m \tau_1$. Since $\Psi \circ T_\gamma^m = \Psi$, its Ψ-image is the conjugate loop $(\Psi\rho)^{-1}(\Psi\sigma'_m)(\Psi\rho)$, which lies in X_g and has class $\pm m$. Suppose $T_\gamma^m D_1 = D_1$, and let $\sigma \subset D_1$ be a path from τ_1 to $T_\gamma^m \tau_1$ (Lemma 7.1). Contractibility of Teich(B) makes the two paths homotopic rel endpoints, so their Ψ-images are homotopic in X_g. But $\Psi\sigma \subset \Delta$, which is contractible, so its class is 0, a contradiction. This is the argument of Proposition 8.1 with the base point moved to τ_1. Hence $T_\gamma^m D_1 \neq D_1$ for every component D_1 and every $m \neq 0$: every orbit of $\langle T_\gamma \rangle$ on $\pi_0(\mathcal{D}_f)$ is infinite. This proves Theorem 1(a). □

Remark 10.1 (a direct topological proof that the lift is trivial)**.** The identity $f^* \circ T_\gamma = f^*$ can also be seen without the fibre computation. Put $s = 1/z - \frac{11}{4}$. Then $f = v + s^2/6$, since $v + \frac{1}{6}(1/z - \frac{11}{4})^2 = v + \frac{121}{96} - \frac{11}{12z} + \frac{1}{6z^2}$ and $v + \frac{121}{96} = 1$. The critical points are $s = 0$ ($z = c_2$, over v) and $s = \infty$ ($z = 0 = a_0$, over $a_1 = \infty$). In this coordinate $s(a_0) = \infty$, $s(a_1) = -\frac{11}{4}$, $s(a_2) = -\frac{7}{4}$, $s(a_3) = \frac{5}{4}$, and the positive numbers $\frac{5}{4} < \frac{7}{4} < \frac{11}{4}$ satisfy $v + s^2/6 = 0, \frac{1}{4}, 1$ respectively.

Put $Y = f - v = s^2/6$. The arc α meets $\hat{\mathbb{R}}$, apart from its endpoints, only at $x_1, x_2, x_3 > v$, i.e. at positive Y. So the interior of α lies in $\mathbb{C} \setminus (-\infty, 0]$ in the Y-coordinate, where $s = \sqrt{6Y}$ has a branch with $\operatorname{Re} s > 0$. Hence $f^{-1}(\alpha) = \alpha_+ \cup (-\alpha_+)$ with $\alpha_+ \subset \{\operatorname{Re} s > 0\}$ a simple arc from $s = 0$ to $s = \infty$, and $J = f^{-1}(\alpha)$ is a Jordan curve through 0 and ∞. The involution $s \mapsto -s$ preserves J and exchanges its two complementary discs Ω, Ω': near its fixed point $0 \in J$ it is a rotation by π which exchanges the two local sides of J.

J meets the positive real s-axis only in α_+, at the three transversal crossings $s_i = \sqrt{6(x_i - v)}$, and $x_3 < 0 < x_2 < \frac{1}{4} < 1 < x_1$ gives $s_3 < \frac{5}{4} < s_2 < \frac{7}{4} < \frac{11}{4} < s_1$. So $\frac{5}{4}$ and $\frac{7}{4}$ lie on opposite sides of J, and $\frac{7}{4}, \frac{11}{4}$ on the same side. Say $\frac{5}{4} \in \Omega$; then $\frac{7}{4}, \frac{11}{4} \in \Omega'$, so $-\frac{7}{4}, -\frac{11}{4} \in \Omega$. Hence $a_1, a_2, a_3 \in \Omega$ and $a_0 \in J$. Now $f^{-1}(N(\alpha))$ is a regular neighbourhood of J, and the collars R_1, R_2 of Lemma 5.3 are parallel to its two boundary curves $\gamma_1 \subset \Omega$, $\gamma_2 \subset \Omega'$. The curve γ_1 separates $\{a_1, a_2, a_3\}$ from $\{a_0\}$, so it is peripheral in (S^2, A), and γ_2 bounds a disc in Ω' containing no point of A. Twists about such curves are trivial in $\mathrm{PMod}(S^2, A)$, so $[\tilde{T}] = 1$, i.e. $f^* \circ T_\gamma = f^*$.

11 Transport to an arbitrary f: proof of Theorem 1(b)

Fix $t \in W$ and a path t_s ($s \in [0,1]$) in W from $t_0 = \frac{1}{4}$ to t. Write $f_s = f_{t_s}$, $B_s = \{0, \infty, 1, t_s, v_s\}$ with $v_s = v(t_s)$, and $A_s = B_s \setminus \{v_s\}$.

Lemma 11.1 (isotopy)**.** *There is a jointly continuous isotopy h_s of $\hat{\mathbb{C}}$ with $h_0 = \mathrm{id}$, fixing $0, 1, \infty$, with $h_s(t_0) = t_s$ and $h_s(v_0) = v_s$.*

Proof. For $|q| \leq \frac{1}{2}$ the map $P_q(z) = z + (1 - |z|)q$ is a homeomorphism of the closed unit disc that is the identity on the boundary and sends $0 \mapsto q$. It maps the disc into itself, since $|P_q(z)| \leq |z| + (1 - |z|)|q|$. It is injective, since $P_q(z) = P_q(z')$ gives $|z - z'| = \big||z| - |z'|\big|\,|q| \leq |z - z'||q|$. It is surjective by invariance of domain, the boundary being fixed. Finally $(q, z) \mapsto (q, P_q(z))$ is a continuous bijection of a compact Hausdorff space, hence a homeomorphism.

Let $\delta > 0$ be less than one third of the minimum, over $s \in [0,1]$, of the mutual distances among $0, 1, t_s, v_s$ (positive by compactness, since $t_s \in W$ keeps these four points distinct and finite). By

uniform continuity choose $0 = s_0 < s_1 < \cdots < s_n = 1$ with $|t_s - t_{s_k}| < \delta/2$ and $|v_s - v_{s_k}| < \delta/2$ for $s \in [s_k, s_{k+1}]$. On $[s_k, s_{k+1}]$ let D_t, D_v be the closed discs of radius δ about t_{s_k}, v_{s_k}; they are disjoint and avoid $0, 1, \infty$. Let Π_s^t be the affine conjugate of P_q to D_t with $q = (t_s - t_{s_k})/\delta$, extended by the identity off D_t (so it moves t_{s_k} to t_s), and similarly Π_s^v on D_v. Set $h_s = \Pi_s^t \Pi_s^v h_{s_k}$. At $s = s_k$ we have $q = 0$ and $P_0 = \mathrm{id}$, so the definitions agree at the junctions; joint continuity follows from that of $(q, z) \mapsto P_q(z)$. □

Lemma 11.2 (lift)**.** *There is a continuous family of homeomorphisms $\tilde{h}_s$ with $f_s \tilde{h}_s = h_s f_0$, $\tilde{h}_0 = \mathrm{id}$, and $\tilde{h}_s = h_s$ on A_0.*

Proof. Write $f_s = M_s Q N_s$ with $Q(z) = z^2$, $M_s = M_{t_s}$, $N_s = N_{t_s}$, by (1). Put $k_s = M_s^{-1} h_s M_0$. Then k_s fixes 0 and ∞, since $M_0(0) = \infty$, $M_0(\infty) = v_0$, $h_s(v_0) = v_s$ and $M_s(\infty) = v_s$. The map $K(s, z) = k_s(z^2)$ on $[0, 1] \times \mathbb{C}^*$ lifts through the covering $Q\colon \mathbb{C}^* \to \mathbb{C}^*$, because the domain retracts to $s = 0$, where $K = Q$. This gives $\tilde{k}_s$ with $Q\tilde{k}_s = k_s Q$ and $\tilde{k}_0 = \mathrm{id}$. By uniqueness of lifts $\tilde{k}_s$ is odd ($z \mapsto -\tilde{k}_s(-z)$ is a lift agreeing with $\tilde{k}_s$ at $s = 0$), hence injective: $\tilde{k}_s(z) = \tilde{k}_s(z')$ forces $z' = \pm z$, and $z' = -z$ would give $\tilde{k}_s(z) = 0$. It is surjective, again by oddness, and it extends by $0 \mapsto 0$, $\infty \mapsto \infty$ to a homeomorphism of $\hat{\mathbb{C}}$. Put $\tilde{h}_s = N_s^{-1} \tilde{k}_s N_0$. Then $f_s \tilde{h}_s = M_s Q \tilde{k}_s N_0 = M_s k_s Q N_0 = h_s f_0$.

On A_0: $\tilde{h}_s(0)$ is a preimage under f_s of $h_s(a_1) = \infty$, whose unique preimage is 0. For $i = 1, 2, 3$, $\tilde{h}_s(a_i)$ is a continuous selection from the two distinct preimages under f_s of $a_{i+1}(s) = h_s(a_{i+1})$, which is not a critical value. These preimages move continuously, one of them is $a_i(s)$, and the selection starts at a_i. So it equals $a_i(s) = h_s(a_i)$ for all s. □

Therefore $\kappa_s = h_s^{-1} \tilde{h}_s$ is an isotopy rel A_0 starting at the identity, so $\tilde{h}_1 \simeq h_1$ rel A_0.

Proposition 11.3. *Let $h = h_1$ and $\tilde{h} = \tilde{h}_1$, and define $h^*[\varphi] = [\varphi h]$ from* Teich(B_1) *to* Teich(B_0)*, and likewise from* Teich(A_1) *to* Teich(A_0)*. Then:*

(i) *$i_* h^* = h^* i_*$ and $f_0^* h^* = h^* f_t^*$;*
(ii) *hence $h^*(\mathcal{D}_{f_t}) = \mathcal{D}_{f_0}$, with components mapping to components;*
(iii) *$h^* \circ T_{h(\gamma)} = T_\gamma \circ h^*$.*

Proof. The first identity in (i) is clear. If $\varphi f_t = F\psi$, then $\varphi h f_0 = \varphi f_t \tilde{h} = F\psi\tilde{h}$, so $f_0^*[\varphi h] = [\psi\tilde{h}] = [\psi h]$ because $\tilde{h} \simeq h$ rel A_0. For (ii), h^* is a homeomorphism with inverse $(h^{-1})^*$. For (iii), $h^*(hgh^{-1} \cdot [\varphi]) = [\varphi h g^{-1}] = g \cdot h^*[\varphi]$, and $hT_\gamma h^{-1} = T_{h(\gamma)}$. □

Hence $T_{h_1(\gamma)}$ preserves $\mathcal{D}_{f_t}$, and every orbit of $\langle T_{h_1(\gamma)} \rangle$ on its components is infinite, by Theorem 1(a).

For an abstract f, take M from Proposition 4.1; it maps B to B_t, since $M(v) = M(f(c_2)) = f_t(M(c_2)) = v(t)$. The map $M^*\colon [\varphi] \mapsto [\varphi M]$ satisfies the same identities as in Proposition 11.3, with $h = \tilde{h} = M$, because $f_t M = Mf$. So $\gamma_f = M^{-1} h_1(\gamma)$ has the properties stated in Theorem 1(b).

Real $t \in I = \left(0, \frac{3-\sqrt{5}}{2}\right)$. The interval I lies in W, and on I we have $v(t) < 0$ (the numerator of $-v$ is a nonzero square and $4t(1-t)(1-2t) > 0$). So the order $v_s < 0 < t_s < 1 < \infty$ persists along the segment from $\frac{1}{4}$ to t. Instead of the isotopy of Lemma 11.1 take $h_s(x + iy) = \phi_s(x) + iy$, where ϕ_s is the increasing piecewise-linear map of $\mathbb{R}$ fixing 0, 1 and everything outside a large interval, with $t_0 \mapsto t_s$ and $v_0 \mapsto v_s$; it is jointly continuous and has all the properties used in Lemma 11.2 and Proposition 11.3. It preserves $\mathbb{R}$, both half-planes, and the intervals between marked points. So $h_1(\alpha)$ has the same crossing pattern, with $\frac{1}{4}$ replaced by t and v by $v(t)$. Pulling back by M gives the description in Theorem 1(b). For non-real t the curve is given as $M^{-1} h_1(\gamma)$ with h_1 the explicit isotopy of Lemma 11.1 along a chosen path in W; every choice of path works. □

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